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ANURAG GARG

DIVYANSH JAIN

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GNNetic: Unpacking Player Decisions in Multiplayer Card Games to Drive Interface Design Improvements and Personalized Skill Enhancement: An Online Rummy Case Study

ANURAG GARG, Data Science Group, Gameskraft, India
DIVYANSH JAIN, Data Science Group, Gameskraft, India

Player decision-making in multiplayer card games presents opportunities to improve human interaction with complex systems. We introduce *GNNetic*, a Relational Graph Convolutional Network for game analysis in card games. By encoding card relationships and game context as a graph, GNNetic frames each discard choice in Rummy (case study) as a graph-classification task learned from expert play, enabling the identification of players' action deviations. We leverage these insights to address core HCI challenges in game design, interface optimization, and player development. GNNetic helps surface recurring patterns of suboptimal play that may indicate design-induced errors, informing iterative refinements to enhance usability and reduce cognitive load during gameplay. Our analysis and summative study demonstrate this potential: UI enhancements informed by GNNetic (*Companion Mode* in practice games) reduce common errors like joker discards by **~31%** and invalid declarations by **~27%**. We also propose *Skill Arena*—a post-game visualization that renders these deviations accessible, providing feedback to foster metacognitive awareness and facilitate skill enhancement. GNNetic achieves a **~15x** reduction in model size with **~6.5%** accuracy improvement over larger baselines. By bridging expert strategy with interface design, this work paves the way for more intuitive UIs and targeted skill enhancement in card games, demonstrating tangible performance gains through interface improvements.

CCS Concepts: • **Computing methodologies** → **Neural networks**; • **Human-centered computing** → **Graphical user interfaces**; • **Applied computing** → *Learning management systems.*

Additional Key Words and Phrases: Game AI, Adaptive Interfaces, Metacognition, Cognitive Support, Relational Graph Convolutional Networks, Game state representation, Player Upskilling.

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1 Introduction

Multiplayer card games rank among the most enduring and widely played social activities worldwide—and with the growth of affordable internet and mobile platforms, they have surged in online and real-money contexts. The global Real Money Skill Games market, encompassing Rummy, Poker, Bridge, and similar skill-based card titles, was valued at USD 17.09 billion in 2023 and is projected to reach USD 19.47 billion in 2024 [23]. Among these, Online Rummy is especially popular in South Asia, with reports estimating over 90 million real-money users in India alone[14, 31].

Authors' Contact Information: Anurag Garg, research@gameskraft.com, Data Science Group, Gameskraft, India; Divyansh Jain, Data Science Group, Gameskraft, India.

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This widespread adoption underscores its significance as both a socially relevant activity and a compelling testbed for research in Human–Computer Interaction (HCI) and Artificial Intelligence (AI).
As a digital skill-based game, Online Rummy presents distinct Human–Computer Interaction (HCI) challenges. Players must make high-stakes decisions under dynamic, partial-information conditions, such as hidden opponent hands, inferred card distributions, and future draws—while tracking evolving game states. They also face high working-memory demands from real-time hand evaluation: rapidly identifying potential melds (Pure Sequences, Sets, Joker Cards) and recalling card properties amid variable play and opponent actions. These demands make Rummy an ideal context for applying Cognitive Load Theory [35, 39], which distinguishes between *intrinsic load* (task complexity) and *extraneous load* (design-induced burden), offering a framework to assess usability and cognitive strain.
A particularly revealing moment of cognitive complexity arises during the *discard* action, which players must execute on every turn. Though seemingly simple, it often triggers critical errors—such as breaking valid Melds or misusing jokers—especially when working memory is overtaxed. These errors not only affect game outcomes but also diminish the player experience. Causes range from cognitive overload and skill gaps to interface designs that amplify extraneous load. Thus, discard decisions provide an ideal focal point for studying how interface design, cognitive scaffolding, and AI-based feedback can support players in reflection.
Existing AI work in digital Rummy has centered on predicting optimal discards (e.g., CardNet [18]), but these approaches overlook why players deviate from ideal moves—whether due to skill gaps or design-induced errors. They also ignore broader game-level context (player count, remaining drops, variant type etc) leading to less accurate models. Building a framework that provides insights to interface enhancements and players demands a fast, efficient model that not only incorporates all decision-influencing factors comprehensively, but must also deliver high accuracy, yet most existing models are large, less accurate, opaque, and ill-suited for near real-time deployment.

To address these limitations, we introduce **GNNetic**, a novel, light-weight Relational Graph Convolutional Network (R-GCN) [36] for analysis of player actions in multi-player card games (Online Rummy as Case Study). GNNetic models expert discard strategies using a graph-based encoding of card relationships and game context. By identifying deviations in player discards, we aim to surface insights into cognitive challenges (leading to interface enhancements) and areas for design and skill improvement (metacognitive support).
Our technical and HCI contributions are:

- **Graph-based Game State Representation:** A novel, unified graph model capturing the entire game state, encoding card-to-card relationships and linking a player's hand to broader game-context factors—in a generic, comprehensive manner.
- **Companion Mode:** which provides contextual gameplay enhancements (e.g., meld grouping, joker cues, score-based prompts) in practice games. Drawing on cognitive apprenticeship [10] and instructional scaffolding principles [20], these features externalize expert strategies and reduce extraneous cognitive load—supporting players in real-time strategic decision-making during practice games. This led to measurable gains—reducing joker discards by **31%** and invalid declarations by **27%** in practice games;
- **Skill Arena:** a post-game visualization tool designed to promote **metacognitive feedback** [19, 34] — by helping players reflect on their decisions, recognize patterns of error, and internalize strategic principles over time. Skill Arena builds on our prior success with *First Drop insights* [21, 28], where structured post-game feedback led to a **2–2.5x** improvement in Drop Conformance - demonstrating the effectiveness of reflective, post-game learning.

Empirically, GNNetic achieves a **~15x** smaller model size and **~6.5%** higher accuracy than CNN baselines, while *Companion Mode* reduces joker discards by **~31%** and invalid declarations by **~27%** in practice games.

By bridging expert-modeled strategy with interface design and **metacognitive support**, this work demonstrates how reducing **extraneous cognitive load** by more intututive UIs and reducing **instrinsic cognitive load** by fostering **self-reflection** leads to measurable performance gains and improved decision-making skills in card games. These games are sometimes also played in real-money contexts. In such scenarios, to ensure fairness and avoid influencing the outcome, it is necessary that these insights are surfaced to the players only post-game.

The remainder of this paper details the core mechanics of Online Rummy and its cognitive demands (Section 2), positions our work within existing literature (Section 3), and then presents the GNNetic-Powered Insights Pipeline, which connects prediction, explanation, and feedback (Section 4). We then describe the GNNetic model architecture (Section 5), followed by model evaluation and results (Section 6), and a discussion on explainability (Section 7). Practical applications are showcased through Companion Mode (Section 8) and Skill Arena (Section 9). We conclude with a summary of findings (Section 10), directions for future work (Section 11), and a reflection on the broader impact of our research (Section 12).

2 Online Rummy: Game Mechanics and Cognitive Demands

Online Rummy is a popular card game enjoyed by 2 to 6 players, characterized by its skill and blend of strategic decision-making. Rummy generally utilizes two standard 53-card Decks, each including a Printed Joker Card per deck. At the start of each deal, each player receives 13 Cards and one card's rank is randomly chosen as Wild Card Joker for the whole Deal (e.g., if 7♦ is chosen, all cards with 7 as rank belonging to any suit become wild card jokers).

Players take turns in a given order, drawing one card—either from the Closed Deck (face-down) or the Open Deck (face-up or discard pile)—and discarding one to the discard pile. The objective is to form valid Melds by organizing cards into Sequences (Pure Sequence or Impure Sequence / Sequence) or Sets. A winning hand must contain at least two sequences, including one Pure Sequence, with all other cards also arranged in valid melds. This cyclical process continues until a player achieves a *Valid Declaration*. A declaration that fails to meet the required structure is considered invalid (also referred to as *Invalid Declaration*).

While general Rummy mechanics underpin most variants, we evaluate GNNetic and report results across all popular Rummy variants (Points, Deals, Pool). However, for deeper analysis, we focus on Pool Rummy, which is the most complex variant of Rummy. In Pool Rummy, players contribute a fixed entry fee to a prize pool. The game continues for multiple rounds, with players accumulating points based on their ungrouped cards after each round. A predetermined point threshold (for example, 61, 101, or 201) based on the pool variant determines player elimination. Players can also opt to *Drop* from a round, incurring a penalty but with a limited number of drops allowed. This multi-round structure, combined with varying point thresholds and drop options, means strategic point management and long-term survival are critical. This heightened contextual complexity makes Pool Rummy an ideal testbed for understanding how interface design impacts **cognitive load** and how personalized feedback can target nuanced skill development.

Fig. 1 illustrates a gametable from an ongoing deal of a Pool 101 Rummy practice game, highlighting a common HCI challenge in real-time hand evaluation. The absence of explicit visual cues for melds, their scoring, and Joker associations around the player's hand increases the cognitive burden of discerning optimal groupings. In this scenario, the player holds a winning hand: a Pure Sequence (8♣, 9♣, 10♣, *J*♣), an Impure Sequence / Sequence: (10♠, *J*♠, *PRINTED_JOKER*), and two Sets: (6♦, 6♥, 6♣) and (3♦, *K*♠, 3♥) — where *K*♠ is a Joker Card. The 2♦ is the single ungrouped card

Fig. 1. A legacy Pool-101 gametable in a practice game, demonstrating the cognitive demands of real-time hand evaluation. Without joker highlighting, meld marking, scoring cues, players must manually group cards—here a winning hand—under time pressure, illustrating the need for interface support.

available for a winning declaration. While this expert player successfully identified the optimal arrangement, such scenarios underscore the difficulties novice or less experienced players face in rapidly evaluating complex hands without clear visual affordances. This example emphasizes the need for improved interface design, informed by our GNNetic insights, to reduce cognitive load and accelerate learning, particularly in identifying melds and potential discards for a swift and accurate declaration during gameplay in practice games. Furthermore, even if players miss optimal arrangements with the improved interface, the post-game Skill Arena visualization provides crucial feedback to facilitate their learning and metacognitive awareness.

3 Related Work

3.1 AI Models for Card Games Analysis

AI in imperfect-information card games—such as Libratus in Poker [5], Pluribus [6], and ReBeL [4]—has produced superhuman agents by combining self-play with deep reinforcement learning. While these agent-centric methods have proven effective at optimizing gameplay in simulated environments, they often fall short when it comes to human-centered applications where interpretability, feedback, and real-time adaptability are crucial. Also unlike such approaches, GNNetic does not aim to play optimally in isolation. Instead, our objective is to develop a framework for evaluating human player decisions against expert strategies—prioritizing real-world applicability, explainability, and alignment with how players naturally think and act.

In the domain of Rummy, prior work such as CardNet [18] use Convolutional Neural Networks (CNNs) to learn discard predictions from image-based representations of the hand and visible table state. While effective, these CNNs are limited by their reliance on spatial grid structures, which do not effectively capture the relational nature of card interactions (e.g., melds, duplicates, joker dependencies). They also produce large, opaque architectures unsuited for user-facing analytics. Moreover, the CNN-based architecture in [18] doesn't allow a seamless way to incorporate crucial context such as player count or overall game context. GNNetic addresses these limitations through a lightweight Relational Graph Convolutional Network (R-GCN) [36], representing each card as a node and encoding domain-specific relationships (e.g., sequence, set, duplicate) as typed edges. This structure enables GNNetic to handle variable hand sizes, joker conditions, and gameplay variants while producing interpretable edge-importance outputs. As a result, the model learns to adapt effectively to evolving in-game situations and align more closely with player strategies.

This focus on structured, interpretable representation aligns with and extends recent research in other card-based domains. For instance, Bertram et al. [3] explore card selection in Magic: The Gathering using structured game encodings and multimodal features. While their work centers on a different game genre, it similarly emphasizes the importance of capturing domain structure for strategic decision-making. GNNetic builds on this philosophy in the context of multiplayer games like Rummy—demonstrating how relational modeling can be applied effectively in these environments. Taken together, these advances point toward GNNetic's potential adaptability to other complex card games, such as Poker or MTG, where cards, player actions, and strategic interdependencies can also be naturally represented as graphs through different relational and temporal layers.

3.2 HCI and Cognitive Load in Strategy Interfaces

HCI research has demonstrated that interface design critically shapes cognitive load and performance in complex tasks. *Cognitive Load Theory* [39] emphasizes how extraneous load-from poorly structured or ambiguous visual information-can overwhelm working memory and hinder task execution. In the context of digital games, HUD (Heads-Up Display) design plays a pivotal role in managing attention and information flow. For instance, Caroux et al. [8] showed that the semantic relevance of HUD elements significantly impacts player performance and experience, emphasizing the need for interface elements that align closely with gameplay objectives. Similarly, Zimmerer et al. [46] demonstrated that multimodal interaction designs in tabletop strategy games can effectively reduce cognitive strain, offering players more intuitive pathways to interpret and act upon in-game information.

While prior HCI studies often rely on controlled experiments or pre-defined UI interventions, few have used large-scale gameplay data to uncover latent interface frictions, especially in multiplayer, imperfect-information games like Rummy. GNNetic fills this gap by surfacing patterns of players' strategic deviations—for instance, repeated mis-groupings around unmarked meld boundaries or joker identification in practice games. These errors, when seen across players and contexts, reveal latent interface frictions that may otherwise go unnoticed.

These insights enable offline interface refinement, helping designers adjust affordances (e.g., meld highlights, clearer joker indicators) to reduce player confusion. Though GNNetic does not implement real-time adaptive scaffolding, it provides a robust foundation for data-driven interface adaptation. This aligns with broader game interface frameworks [41] and principles from adaptive tutorial systems [1], supporting the design of interfaces that evolve based on player needs. These refinements are instantiated through *Companion Mode*, which operationalizes expert-modeled insights into actionable interface cues—minimizing extraneous load and scaffolding decision-making in practice games.

3.3 Computational Support for Skill Acquisition

Adaptive feedback and Intelligent Tutoring Systems (ITS) have long enhanced learning in structured domains (e.g., physics, language) by diagnosing errors and offering personalized guidance [2, 33]. In gaming, early systems like MiBoard—a multiplayer version of the iSTART board game—used collaborative play and motivational mechanics to support inquiry-based learning [12]. Similarly, Projective Replay Analysis has been proposed as a method to help align player behaviors with educational goals by using in-game replays as reflective tools [11, 24]. While such systems translate gameplay data into feedback, they often fall short in open-ended, multiplayer settings like Rummy, where strategic decisions under evolving information hinge on both hidden (opponents' hands, future draws) and visible context (player count, rounds played, scores, drops remaining).

GNNetic fills the above gap by using an expert-trained R-GCN to automatically pinpoint subtle strategic deviations from expert play. The objective of post-game in-app *Skill Arena* is to deliver metacognitive feedback - highlighting instances where contextual cues led to suboptimal choices and suggesting alternative strategies. This approach explicitly aims to foster metacognitive awareness and self-regulation in players [18, 36], thereby cultivating transferable decision-making skills grounded in real gameplay by enabling players to reflect on and understand their own cognitive processes and errors. *Skill Arena* builds on our prior work [21, 28] in which players received feedback on their First Drop decisions after the conclusion of their game. Players were shown whether expert players would have chosen to drop or play, given the same hand and context, along with supporting explanations. This led to a **2–2.5x** improvement in Drop Conformance[1]. These outcomes are not only encouraging but also theoretically grounded. Structured, post-game feedback can function as a form of scaffolded learning [20], enabling players to improve through targeted guidance situated in their Zone of Proximal Development (ZPD). This aligns with Ericsson's theory of expertise development [17], which emphasizes feedback-rich, effortful practice as a pathway to mastery. By encouraging reflection and adjustment, *Skill Arena* also promotes self-regulated learning [47], helping players to monitor and refine their decision strategies over time.

3.4 Explainable AI and Interface Feedback

Interpretability is crucial when AI predictions are intended for end-users, particularly in cognitively demanding settings like strategy games. Doshi-Velez and Kim [13] advocate for a principled science of interpretability in machine learning, highlighting the tension between model fidelity and human comprehensibility. While their framework is widely cited, applying these principles to fast-paced, multiplayer card games remains relatively underexplored.

In the domain of games, Gero and Chilton [22] investigate how users form mental models of AI agents, emphasizing the need for explanations that are intuitive and behaviorally aligned. However, their work centers on interpreting AI behavior in general rather than surfacing feedback tied to a user's actions. Liao et al. [32] and Wang et al. [43] contribute foundational design guidance for explanation interfaces, stressing that intelligibility must align with users' goals and decision contexts. Yet, many of these studies are evaluated in controlled or single-user settings, which limits their direct applicability to dynamic multiplayer settings like Rummy.

GNNetic complements this line of work by embedding interpretability into gameplay semantics. While model-level introspection (e.g., visualizing graph attention or message-passing weights) is still possible, GNNetic prioritizes player-centered interpretability—surfacing insights in terms players naturally understand, such as meld formation, optimal grouping, hand minimization, and valid declarations [15, 16]. This allows the system to deliver feedback that is visual, intuitive, and immediately actionable, aligning closely with HCI goals of supporting decision-making under cognitive load.

3.5 Graph Neural Networks in Decision Analysis

Graph Neural Networks (GNNs) excel at learning from relational, non-Euclidean data [30]. In gaming, homogeneous Graph Convolutional Networks (GCNs) have been applied to a variety of tasks by aggregating each node's neighborhood uniformly. For instance, Keller et al. [29] applied GNNs to the board game Hex, demonstrating their effectiveness in capturing long-range dependencies in game states. In Multiplayer Online Battle Arena (MOBA) games like Dota 2, GNN-based approaches have been proposed to model player compatibility and predict game outcomes by

[1]A measure of how often players choose to drop a hand when they should - i.e., when expert players would also drop in the same situation. It reflects whether players are making smart, timely decisions to exit weak hands.

representing players and their interactions as graph structures [37]. In the realm of collectible card games, Bertram et al. [3] explored the use of generalized card representations to improve AI-based deck building in "Magic: The Gathering," highlighting the potential of GNNs in modeling complex relationships between cards. These approaches demonstrate GNNs' power for structured reasoning, but they treat every edge with the same transformation, failing to capture why different relationships matter.

By contrast, Relational GCNs (R-GCNs) [36] implement relation-specific message passing, assigning distinct weight matrices to each edge type. This fundamental difference allows them to better capture varied data relationships. In GNNetic, we define edges for sequences, sets, and duplicates, enabling the model to learn how each relationship between cards uniquely influences discard decisions. This relational reasoning is crucial in Rummy's imperfect-information context, where the type of card connection—rather than mere adjacency—drives human strategy. GNNetic thus pioneers fine-grained, post-game behavioral analysis tied to HCI objectives, delivering both the efficiency and accuracy required for scalable, human-centered game analytics.

While earlier work in AI and HCI has made progress in prediction modeling, gameplay feedback, or interface design individually, few approaches bring these elements together into a single, coherent system. There is growing interest in frameworks that not only identify expert decisions, but also help players understand and learn from their deviations in a meaningful way. The GNNetic-powered insights pipeline, introduced in the next section, contributes to this space by supporting a feedback-oriented system that connects prediction, explanation, and user support—linking gameplay modeling to both learning and interface enhancement.

4 GNNetic-Powered Cognitive Support Pipeline: From Strategic Prediction to Reflective Feedback

Fig. 2 outlines the GNNetic-powered insights pipeline, which integrates predictive modeling with cognitively grounded feedback to support post-game player learning and interface refinement. Starting from raw gameplay logs, the system constructs a graph-based representation of each game state, encoding card relationships and contextual features. This serves as input to a lightweight R-GCN trained to model expert discard behavior. The predictions are contextualized by a best-grouping module that identifies valid melds, point-minimizing configurations, and valid declarations.

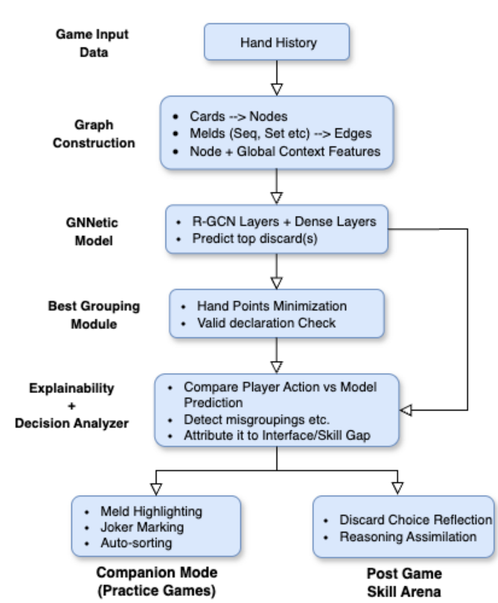

Fig. 2. GNNetic-Powered Insights Pipeline–A unified flow from discard prediction to player feedback (post game), where GNNetic drives expert modeling, interpretable insights, and targeted support via Companion Mode and Skill Arena.

An explanation layer compares player moves to expert-like predictions, surfacing deviations and attributing them to cognitive, skill-based, or interface-related factors. These insights drive two downstream interventions: *Companion*

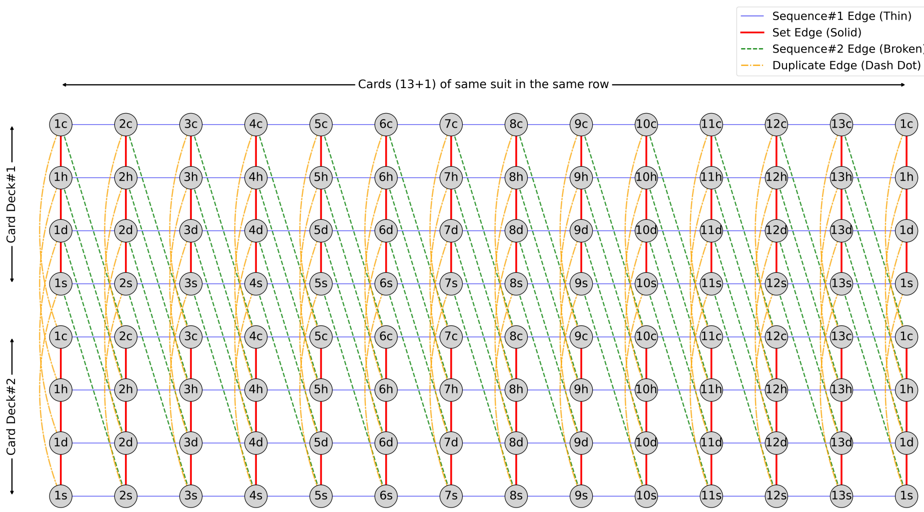

Fig. 3. Graph representation for the game of Online Rummy with cards as nodes and different edges capturing relationships between them.

Mode, which embeds contextual cues in practice games to reduce cognitive load, and *Skill Arena*, which delivers post-game visual feedback for reflective learning.

The effectiveness of both these systems relies heavily on the quality and interpretability of GNNetic's recommendations. Accordingly, the next sections of the paper comprehensively examine GNNetic's architecture, training, evaluation, and explanatory power-before exploring how its insights translate into practical design enhancements and metacognitive support.

5 GNNetic

5.1 Overview

GNNetic introduces a novel graph-based representation for Online Rummy, modeling not just the visible game state (cards and their relationships) but also the overall game context. Each card is represented as a node in a heterogeneous graph, with typed edges capturing domain-specific relationships such as sequences, sets, and duplicates. Additionally, global signals—such as the number of players at the table, how many are still playing, draw/discard history, and game progression indicators—are integrated as contextual features. This holistic encoding allows GNNetic's Relational Graph Convolutional Network (R-GCN) to learn discard strategies that mirror expert decision-making, accounting for both local card structure and global gameplay dynamics. The result is a compact yet expressive architecture that enables more accurate and interpretable post-game feedback across Rummy variants.

5.2 Graph Construction

- Nodes (112 total):
  - 104 non-joker cards (2 decks × 52 ranks).
  - 8 extra Ace nodes (4 suits × 2 decks) to model the dual role of Aces—allowing them to connect both at the start (A–2–3) and end (Q–K–A) of sequences.

- Node Features:
  - The information within each node is encoded using node features which collectively represent the visible game state available to the player.
  - Table 1 lists different node features that are broadly classified based on their role in representing the game state.
  - To provide a more in-depth understanding of node features, Fig. 5 illustrates how these features map to the overall game state for a sample hand.
- Edges (4 types): The relationships between different nodes are captured through various edge types:
  - Sequence #1 (same deck): consecutive ranks, same suit (e.g., 5♠ – 6♠)
  - Set: same rank, different suits (e.g., 5♣ – 5♥)
  - Duplicate: identical card across decks (e.g., two 5♣ nodes). Kept separate to learn distinct roles for identical cards—such as forming a pure sequence with one and holding the other for future meld flexibility.
  - Sequence #2 (cross-deck): consecutive ranks, same suit across decks.

This construction (Fig. 3) remains invariant not only across different actions in Rummy (like Pick or Drop) but also across Rummy variants (like Points, Deals), ensuring generalizability.

5.3 Network Architecture

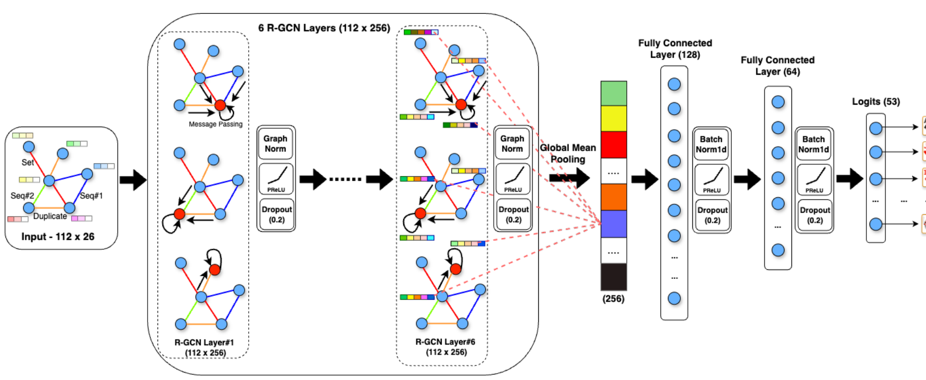

Fig. 4. GNNetic Model Architecture combining Relational GCN layers with dense layers to capture card-to-card and game-context dependencies for expert-level discard prediction.

We stack six R-GCN layers (Fig. 4), each with:

- Relation-specific message passing (distinct weight matrices per edge type)
- Parametric ReLU activation for adaptive nonlinearity [25]
- Graph normalization for stable training [7]
- Dropout (20%) to prevent overfitting

Six layers allow each node to aggregate information up to six hops—sufficient to capture melds of up to five cards, in alignment with the meld length rule of the game of Rummy.

A global mean pooling operation then condenses all node embeddings into a single graph representation, which passes through two fully connected layers, followed by a 53-way softmax classifier (respectively to 52 standard cards + printed joker). These fully connected layers integrate card-level relational embeddings with global gameplay context (e.g., drop availability, elimination risk), enabling the model to capture non-linear strategic factors that simple node comparisons cannot.

Algorithm 1 Graph Construction for Rummy as shown in Fig. 3

```
function add_seq_edge(G, v_i, v_j)
    G(v_i, v_j) ← 0
end function

function add_set_edge(G, v_i, v_j)
    G(v_i, v_j) ← 1
end function

function add_same_card_edge(G, v_i, v_j)
    G(v_i, v_j) ← 2
end function

function add_seq_2_edge(G, v_i, v_j)
    G(v_i, v_j) ← 3
end function

for each vertex v_i in G(V, E) do
    rank_{v_i} ← rank(v_i)
    suit_{v_i} ← suit(v_i)
    rank_{v_i,neighbour} ← next_rank(v_i)
    v_i_neighbour ← get_vertex(rank_{v_i,neighbour}, suit_{v_i}, is_same_deck = True)

    add_seq_edge(v_i, v_i_neighbour) //Add sequential edge between nodes from the same deck

    v_i_neighbour ← get_vertex(rank_{v_i,next}, suit_{v_i}, is_same_deck = False)
    add_seq_2_edge(v_i, v_i_neighbour) //Add sequential edge between nodes from both decks

    v_i_neighbour ← get_vertex(rank_{v_i}, suit_{v_i}, is_same_deck = False)
    add_same_card_edge(v_i, v_i_neighbour) //Add duplicate edge between nodes from both decks

    for each suit in {c, h, d, s} do
        if suit ≠ suit_{v_i} then
            v_i_neighbour ← get_vertex(rank_{v_i}, suit, is_same_deck = True)
            add_set_edge(v_i, v_i_neighbour) //Add set edge between nodes from both decks
        end if
    end for
end for
```

We treat *discard* prediction as a graph-level supervised classification problem. This formulation, combined with *Label Smoothing* [40], mitigates overconfidence under the *one-label constraint* and delivers both high accuracy and compactness for real-world deployment. The algorithm (Algorithm 1) outlines how the graph in Fig. 3 is being constructed.

Table 1 presents a selection of key node features designed to capture the diverse factors that influence the discard decisions of highly skilled players. These features can be broadly classified based on their role in representing the game state.

Fig. 5 illustrates the full 112-node graph with encoded features for a sample 14-card hand for a Pool-6p game. Each node's feature vector—defined in Table 1—is computed for all cards in the deck, and for clarity, we annotate the specific feature values for the sample hand directly alongside the corresponding entries.

Table 1. Categories of Input Node Features and Sample Values. Breakdown of card-level feature categories used in the graph input, illustrating how a representative hand (Fig. 5) is encoded for GNNetic's learning.

Card Level - Fundamental Features *Card Rank*: Rank of the card (1-13) *Card Suit Category*: Suit category of the card (*0, 1, 2, 3* respective to *c, h, d, s*) *Card is Joker*: 1 if the rank of the card is the same as the rank of the Joker, 0 otherwise (e.g. in Fig. 5, *K*♠ - *13s*) *Card Adjacent to Joker*: 1 if the card is neighbor of the declared joker, 0 otherwise (e.g. 12♥ and 1♥ are neighbors to the declared joker *K*) *Card Points*: Points contributed by the card towards hand (0-10, 0 if the card is joker) (e.g. *K*♠ - 13s, being joker, contribute 0 points and all other non-joker cards with value as their rank, maximum being 10) *High Card*: 1 if card points ≥ 10, 0 otherwise (e.g. 10♠, *J*♠ are high cards with points >= 10)
Card Level - Positional Features *In Hand*: 1 if the card is in the hand of the player, 0 otherwise (e.g. in Fig. 5, the first 14 columns are set to 1 except printed joker - *j*; all others being 0) *Picked by Next Neighbor*: 1 if the card has been picked by the next player (from Open Deck), 0 otherwise (e.g. *J*♠ - *11s*) *Discarded by Next Neighbor*: 1 if the next player has discarded the card, 0 otherwise (e.g. *Q*♥ - *12h*) *Picked by Previous Neighbor*: 1 if the card has been picked by the previous player (from Open Deck), 0 otherwise (e.g. 1♦) *Discarded by Previous Neighbor*: 1 if the previous player has discarded the card, 0 otherwise (e.g. 1♦) *In Discard Pile*: 1, if any player has discarded the card, 0 otherwise (e.g. 9♠)
Player's Hand Level Information *Total jokers in hand*: 0-9 (Two decks) (set to 2 - jokers in hand being printed joker - *j* and *K*♠) *Total printed jokers in hand*: 0-2
Player's Game Level Features *Player Drops Left*: In the case of a pool game, how many First Drops can a player still do? (4 - Player is on 20 points in 101-pool game and can drop 4 more times with a penalty of 20 points) *Can get eliminated*: If the game ends now, will the player get eliminated from a pool game? *Current Game Score Ratio*: Player's total points (including the ongoing deal) relative to the game's score limit. (set to 0.22 - (20+2)/101) *Player Turn*: Turn number at the game level
Global Game Context *Declared Joker Rank*: The rank of Joker in the game (13 - as *K* is the declared joker) *Game Variant*: Type of Game variant (Pool 201, Pool 101, etc.) *Players joined*: Total number of players who joined the game in the beginning (2-6) (3 players had joined that deal) *Players still playing*: Total number of players still playing on the table (2-6) (3 players are still playing on the table)

Cards	10♠	J♠	Joker	8♣	9♣	10♣	J♣	6♦	6♥	6♠	3♦	K♠	3♥	2♦	..	J♦	9♠	Q♥	1♦	..
Card Level - Fundamental Features																				
Rank	10	11	-	8	9	10	11	6	6	6	3	13	3	2		11	9	12	1	
Suit	3	3	-	0	0	0	0	2	1	3	2	3	1	2	..	3	3	1	2	..
Points	10	10	-	8	9	10	10	6	6	6	3	0	3	2	..	10	9	10	10	..
Is_Joker	0	0	-	0	0	0	0	0	0	0	0	1	0	0	..	0	0	0	0	..
Is High Card	1	1	-	0	0	1	1	0	0	0	0	0	0	0	..	1	0	1	1	..
Adjacent to Joker	0	0	-	0	0	0	0	0	0	0	0	0	0	0		0	0	1	1	
Card Level - Positional Features																				
In Player's Hand	1	1	-	1	1	1	1	1	1	1	1	1	1	1		0	0	0	0	
Picked by Next Neighbor	0	0	-	0	0	0	0	0	0	0	0	0	0	0		1	0	0	0	
Discarded by Next Neighbor	0	0	-	0	0	0	0	0	0	0	0	0	0	0	..	0	0	1	0	..
In Discard Pile	0	0	-	0	0	0	0	0	0	0	0	0	0	0	..	0	1	1	0	..
Picked by Previous Neighbor	0	0	-	0	0	0	0	0	0	0	0	0	0	0	..	0	0	0	1	..
Discarded by Previous Neighbor	0	0	-	0	0	0	0	0	0	0	0	0	0	0		0	1	0	0	

Global Features

Player's Hand Level Features	
Total jokers in hand	2
Total Printed Jokers Count	1

Player 's Game Level Features	
Turn Number	20
Drops Left	4
Elimination possible	0
Current Game Score Ratio	0.22

Common Game Level Features	
Declared Joker Rank	13
Game Variant	101
Players joined	3
Players Playing	3

Fig. 5. Structured Representation of a Sample Hand in a Pool-6P game–depicting how GNNetic encodes card-level and game-context features for a player's hand to support expert-informed discard predictions and insight generation.

5.4 Training Methodology

We define a **High-Skilled Player (HSP or expert)** as a player who has played at least 100 Pool-6p cash games on high-value tables (buy-in ≥ $Rs.1000$) within 30 days, achieving a differential end score ≥ 3 and a win rate of at least 20% (following [18]). We considered all historical Pool-6p games played by these players on our platform, resulting in a dataset of ~2,500 players, covering 250,000+ deals and 500,000+ discard moves. For each move, we capture all the information that was visible to the player such as their hand, opponent discarded/picked cards, declared joker and game specific

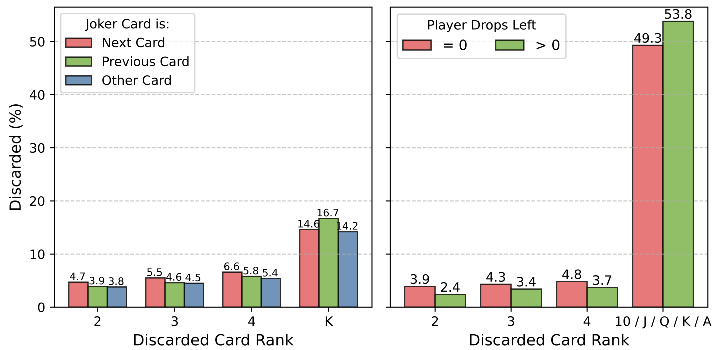

Fig. 6. Patterns in Player Discard Behavior–(Left) Players discard cards adjacent to jokers more frequently, indicating attention to meld potential and flexibility. (Right) Discard strategies shift with remaining drops—when no drops are left, players increasingly discard low-value cards (2–4) while retaining higher-value cards (10–A), reflecting risk-tolerant play in critical rounds.

information such as current score-board, players on the table etc. The target attribute consisted of which card was discarded by the player for that move. To mitigate bias and ensure generalizability to unseen players, we randomly assigned 20% of players to both validation and test sets, ensuring no overlap.

Table 2. HSP Strategic Discard Behavior Patterns. It shows how often HSPs perform each action.

Discard Behavior	%
Avoids elimination and continues (when possible)	99
Avoids forming partial meld with next player's picked card	87
Avoids forming partial meld with previous player's picked card	83
Doesn't break own partial sequence	79
Forms partial meld with next player's discarded card	62
Is a duplicate card	34
Already discarded by next player	15

A deeper analysis of their gameplay behavior reveals that discard decisions are influenced by the 14 cards in hand and various external factors. Some examples:

- **Risk-Driven Discard Strategy**: As elimination nears in a Pool game (Drops Left = 0), HSPs are more open to discarding low value point cards while retaining high-point cards if they contribute to partial melds Partial Meld (Fig. 6).
- **Duplicate and Joker-Adjacent Discards**: HSPs prefer to discard duplicate cards (Table 2) and cards adjacent to the declared joker. They show an even stronger tendency when the declared joker is a high rank card (Fig. 6).
- **Opponent-Aware Decision-Making**: HSPs track opponent discards and picks, avoiding discarding neighboring cards to those picked by the next player from Open Deck Pile while discarding neighboring cards more frequently when the next player has already discarded similar ones (Table 2).
- **Strategic Use of Previous Neighbor's Discards**: HSPs are more likely to discard neighboring cards if their previous neighbor has already picked similar ones from Open Deck Pile (Table 2).

These strategies and those listed in Table 2 help HSPs optimize discards, mitigate risk, and minimize opponent advantages.

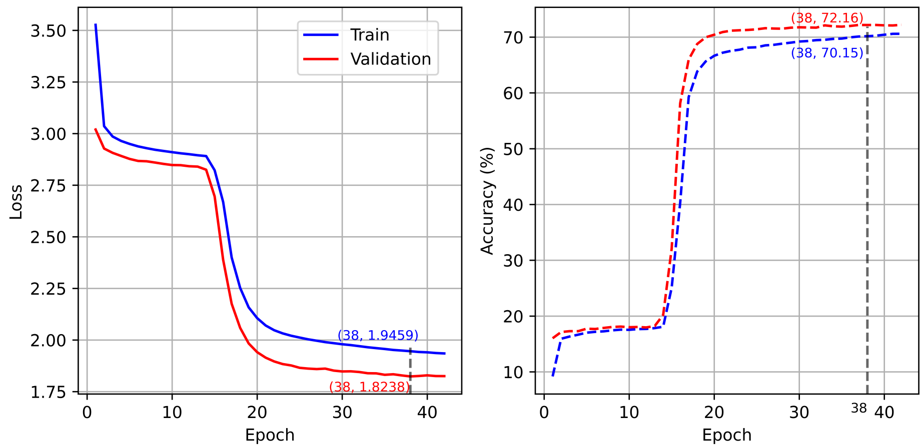

Fig. 7. Accuracy and Loss vs Epoch

Previous studies have not fully accounted for the game specific nuances influencing discard decisions. Incorporating these specific attributes into the modeling process is essential to achieve more accurate predictions of the optimal discard action.

6 Model Evaluation and Results

6.1 Model Performance

Since it is a multiclass classification problem, *Categorical cross-entropy* is used as the loss function. Training is conducted over 100 epochs with a batch size as 256 using *Adam Optimizer*. Early Stopping is implemented to prevent overfitting, and training is halted if the validation loss does not decrease for more than 4 continuous epochs. Fig. 7 shows training and validation datasets' loss and accuracy plots. As the number of epochs increases, the loss converges. Notably, we observe a sharp improvement in validation accuracy around Epoch#15. Upon closer analysis (inspecting predictions before and after this point), this inflection point corresponds to the model learning to preserve valid sequences, thereby making more strategic discard choices. This transition reflects a meaningful internalization of core gameplay patterns. Training remains stable beyond Epoch#40, and the final model is selected at Epoch#38—where validation loss reaches a consistent minimum.

Table 3 presents the Top-1 accuracy of different models for the Pool-6P game type on the test set. The *XGBoost* model was selected as a baseline known for its robustness and performance on tabular data. It relied on handcrafted domain knowledge features as per the commonly known strategies of the game. XGBoost yielded the lowest accuracy, highlighting the limitations of traditional feature engineering.

CNN-based models (*CardNet*, *CardNet-1*) outperformed the tree-based models, demonstrating their efficacy in learning card relationships. We exclude the *CardNet-L* model as proposed in [18] for our comparison. While look-ahead modeling is feasible in 2-player games, where a single opponent makes future states more predictable, it becomes impractical in multi-player settings. The exponential growth of possible game trajectories, increased hidden information, and unpredictable player actions make accurate forecasting difficult. Additionally, relatively shorter game durations and infrequent turns hinder look-ahead modeling, as significant state changes occur between player turns, reducing future predictions' reliability. Notably, a graph-based model achieved the

Table 3. Comparison of Models and Their Performances for a Pool-6p Variant of Rummy.

Model	**Description**	**Accuracy (%)**
XGBoost	Multi-class classification model using domain knowledge features, including current hand meld state (pure sequence, impure sequence, sets, hand points, partial sequence count, joker rank) and changes upon discarding a card. The discarded card serves as the ground truth label.	49.28
CardNet[1]	As described in [18].	67.75
CardNet-1	Multi-input neural network incorporating CardNet as a CNN block, as detailed in [28]. Dense layer features include game variant, drops left, opponent drops left, and turn number.	69.32
R-GCN	Relational Graph Convolutional Network (R-GCN) where each card is represented as a node (112 nodes total) with edges capturing sequence, duplicate, and set relationships. The architecture includes four relational graph convolutional layers with graph normalization, PReLU activation, and 20% dropout. A global mean pooling layer aggregates node features, followed by two dense layers leading to a softmax classification output. Each node is represented by 11 descriptive features. *(# params 228,341)*	66.58
R-GCN-global	*R-GCN* with additional global-level game information (15 features). *(# params 854,460)*	67.38
R-GCN-combined	*R-GCN* with additional node-level game information (15 new features, total 26 per node). *(# params 861,115)*	70.04
R-GCN-combined-LS	*R-GCN-combined* with *Label Smoothing* applied, assigning 0.74 probability to the top class and distributing the remaining 0.26 across 13 other classes (or other 13 cards in hand). *(# params 862,139)*	70.99
Res-R-GCN-combined-LS	*R-GCN-combined-LS* with residual skip connections [26], where each layer also receives the previous layer's input as an additional input. *(# params 3,298,747)*	70.11
Large-R-GCN-combined-LS (GNNetic)	*R-GCN-combined-LS* with the number of convolutional layers increased from 4 to 6. *(# params 1,388,477)*	72.16

[1] when reproduced on our dataset for Pool-6p

highest performance. Experimental results indicate that incorporating game-specific features at the node level was more advantageous than global features, suggesting that fine-grained contextual information enhances predictive performance when kept alongside the local features.

We also employ Label Smoothing [40] to address the one-label constraint and overconfidence of standard cross-entropy [44]. Instead of using a hard one-hot target, we allocate probability α to the expert's chosen discard and distribute the remaining $(1 - \alpha)$ uniformly across the other 13 cards. We tested α values between 0.6 and 0.9 and found $\alpha = 0.74$ best balanced confidence and flexibility. With this setting, strong hands (e.g., those with 2+ melds and low hand points) retained peaked predictions, while weaker hands (e.g., no melds and few jokers) surfaced multiple plausible discard candidates. Although this adjustment primarily aimed to improve the *Threshold Accuracy* (discussed in next section), it also resulted in a modest gain in Top-1 accuracy, confirming that soft targets enhance generalization in discard prediction.

By contrast, adding *skip connections* did not yield significant gains. This is likely because the network is relatively less deep (e.g. 4 layers), and message passing alone effectively propagates information, rendering skip connections unnecessary. Finally, our best-performing model, GNNetic, increases depth from four to six R-GCN layers, indicating that deeper architectures can more effectively capture complex relationships in graph-structured data. These models are considerably more accurate (~6.5%) and highly efficient (Only ~1.3 Million parameters - ~15x smaller) than the best-known CNN-based models.

To further understand the practical significance of this accuracy improvement from a player and interface enhancement perspective, we conducted a move-level comparison between GNNetic and the CNN-based discard model using a confusion matrix. GNNetic surfaced 8.34% more actionable insights while reducing only 4.74% of insights that would have previously been shown—resulting in a net gain of 3.60% in high-quality, strategic feedback. These results underscore the advantage of deep graph neural networks, augmented with label smoothing, in capturing complex dependencies within structured card game environments.

As a final step in selecting the best architecture, we also considered integrating attention mechanisms into the graph model. To begin, we replaced the boolean node feature *In Discard Pile* (Table 1) with the actual turn number of each card's discard to better encode discard history. However, this modification did not yield a statistically significant improvement in accuracy. Given the added complexity and limited benefit, we did not pursue attention mechanisms further for this aspect. We thus selected the 6-layer R-GCN with Label Smoothing—referred to as GNNetic—as our final model.

6.2 Threshold Accuracy

Continuing with the model evaluation, we now examine it from even more nuanced angle, aligned with our broader goal of supporting meaningful player learning and decision-making. In real gameplay—as well as in GNNetic's predictions—there are often multiple viable discard options, particularly during early turns or when players hold partially-formed hands. In such situations, the model assigns comparable probabilities to several cards. Relying solely on strict *Top-1 Accuracy* may unfairly penalize these strategically sound decisions, even though they align with expert reasoning. If such moves are flagged as mistakes in post-game feedback (*Skill-Arena*), it can confuse the player, erode trust in the system, or reduce confidence—especially when the difference between options was marginal.

To address this, we adopt *Threshold Accuracy*, a more forgiving evaluation metric proposed in prior work [18]. This involves normalizing the card discard probabilities in the range of 0-1 by dividing them by the highest prediction and then considering all cards above a threshold as valid discards. Accuracy can be measured at different thresholds and plotted to determine the operating

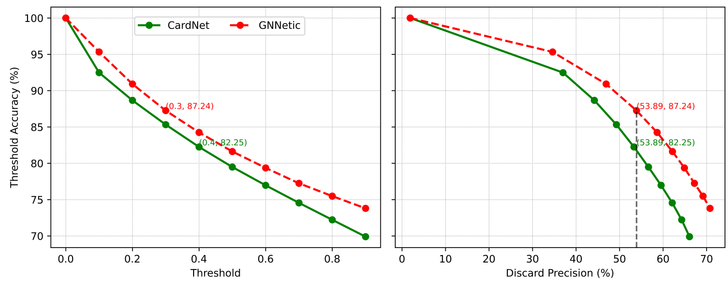

Fig. 8. Threshold-wise Accuracy Comparison of CardNet and GNNetic–GNNetic consistently outperforms CardNet across discard precision and threshold settings, indicating more expert-aligned and dependable discard predictions.

threshold. Fig. 8 shows the comparison of threshold accuracy between GNNetic and CardNet. It is visible that GNNetic has higher Threshold Accuracy at all operating thresholds.

Table 4. Accuracy: Pool-6P

Accuracy	(%)
Top-1	72.16%
Top-2	89.19%
Top-3	94.85%
Threshold	87.24%

However, when deciding the threshold value, CardNet [18] doesn't consider the number of false positives (cards that are valid discards as per the model, but they are not). Theoretically, a random model that gives equal probabilities to all the cards would achieve perfect value for Threshold Accuracy irrespective of the chosen threshold. Further, a lower threshold can be chosen to boost the Threshold accuracy. To balance this, we define *Discard Precision* - a measure of how often the discarded card aligns with the model's suggested choices, for a given threshold. An ideal model that only predicts the player's discarded card would achieve a *Discard Precision* of 1.

Discard Precision is calculated and plotted in Fig. 8 for different thresholds. As expected, if higher Discard Precision is desired, the Threshold Accuracy decreases since fewer cards are predicted as valid discards. This can help pick an optimum threshold to balance both - *Threshold Accuracy* and *Discard Precision.* In addition to Threshold Accuracy, we report different accuracies for GNNetic in Table 4. The results demonstrate that GNNetic effectively captures HSP discard patterns, with a high percentage of actual discards appearing within the Top-2 and Top-3 predicted cards.

6.3 Edge and Feature Importance

To evaluate whether GNNetic internalized different inter-card relationships during training, we conducted an ablation study at both the edge and feature levels (Table 5). At the edge level, we selectively removed one edge type at a time (*Sequence#1*, *Set*, *Duplicate*, *Sequence#2*) and measured the resulting drop in test accuracy. This approach, inspired by permutation importance, avoids retraining and identifies which learned relationships the model relies on most.

Sequence#1 edges—capturing in-hand sequential relations—had the largest impact, highlighting the importance of valid sequences in meld formation. This aligns with the performance jump observed in Fig. 7, where the model began consistently preserving sequences around *Epoch#15*. Set edges were also important, with their removal disrupting groupings of same-rank cards. *Duplicate* and *Sequence#2* edges had lower impact, suggesting that exact duplicates and cross-deck sequences are less critical for discard decisions.

At the feature level, Joker-related features showed the highest influence on model performance, reflecting their central role in Rummy strategy. Additionally, when the First Drop opportunities are exhausted, the model relies more on opponent behavior and variant-specific features—indicating that players shift focus to survival and opponent modeling in high-stakes late-game scenarios (Table 5).

Table 5. Combined Impact of Edge and Feature Removal on Accuracy

Type of Change	Feature Removed	Accuracy Drop (%)	
		Drops Left = 0	Drops Left > 0
Edge Type	Sequence#1	50.56	56.91
	Set	26.02	26.47
	Duplicate	23.17	24.33
	Sequence#2	5.23	6.78
Feature	Joker	12.95	14.90
	Opponent	2.11	1.61
	Game-level	3.30	1.22

This further underscores that GNNetic's decisions are grounded in interpretable, game-relevant concepts (e.g., melds, jokers, phase of play). This strengthens its suitability for downstream, player-facing applications like *Skill Arena* and *Companion Mode*, where insights must be cognitively plausible and reflect meaningful gameplay logic without overwhelming users with opaque reasoning.

6.4 Performance Across Different Aspects of Game

GNNetic's predictions must be robust to generate cognitively plausible and interpretable insights for player-facing use. To ensure this, we further evaluate the model across key attributes—such as its game-level understanding, move-level accuracy, and its alignment with expert strategies. We also analyze instances where the model diverges from expert behavior to understand its limitations and edge cases.

Game-Level: We also evaluated the model across key attributes such as game variant, player drops left at the deal's start, and players on the table (Table 6). As the number of drops left with the player increases, the model accuracy also increases. A key reason for that is when the players have drops left with them, they only play stronger hands and drop the weaker ones (No Pure Sequence and Joker). In stronger hands, where more cards are already arranged in melds, the discard decision becomes more straightforward, as the number of viable discard options is reduced, making the choice less ambiguous. Similarly, when the number of players on the game table reduces, players opt for riskier moves, such as not discarding high-value cards to optimize points but rather optimizing to increase the number of winning permutations.

Table 6. Model performance across different game-level attributes

Game Variant	Acc. (%)
61	70.00
101	71.09
201	72.54

Player Drops Left	Acc. (%)
0	69.39
1	73.05
2	74.43
3	74.56
4	74.88

Players Joined	Players Playing				
	2	3	4	5	6
3	70.17	70.30	–	–	–
4	69.89	71.05	70.72	–	–
5	73.66	72.88	73.20	71.19	–
6	72.37	72.69	74.02	74.09	73.49

Move-Level: We further looked at the model's performance at the move level. Table 7 shows the model's accuracy for the player card melds' type (the 14 in-hand cards). We can see how the model becomes better at identifying the optimum card to discard as the player's hand gets better (increase in number of sequences). This aligns with the real world as the more melds players have, the fewer cards are available for discard, making this decision more straightforward. On the flip side, when the player has a weaker hand (no pure sequence and joker), the player optimizes for making the first sequence, and the number of cards that can be discarded is higher, making this decision harder.

HSP-strategies: Key HSP strategies, such as discarding duplicates, cards adjacent to the joker, and cards unlikely to be picked by the next player (i.e., adjacent to their discards) were looked at. Conversely, HSPs avoid discarding cards adjacent to those recently picked by the next player, as these are more likely to be of interest to them. GNNetic effectively learned and replicated these strategic behaviors.

Table 7. GNNetic Accuracy by Meld Composition and Card Context (Pool-6P) – As hand strength increases (e.g., more pure sequences, jokers), GNNetic's Top-1 discard prediction improves. Accuracy also varies with expert-relevant card features such as duplicates, joker adjacency, and proximity to neighbor actions, highlighting the model's nuanced grasp of strategic contexts.

(a) by player cards (14) melds type

Pure Seq. Count	Joker Count 0	1	2	≥3
0	63.05	67.24	67.21	68.11
1	67.06	73.95	74.56	72.42
2	75.63	77.50	76.15	71.17
3	81.83	78.12	78.34	80.95
4	84.21	83.33	50.00	–

(b) by HSPs Discard behavior

Player has a	Cases (%)	Acc. (%)
Duplicate card	45.23	71.9
Card adjacent to Joker	87.16	70.96
Card adj. to next picked card	10.83	70.78
Card adj. to next discarded card	54.07	70.83

Model–Expert Divergence: To investigate the sources of error in the model's predictions, we conducted a detailed analysis of selected player moves in collaboration with domain experts. The objective was to identify strategic patterns where the model's behavior deviated from that of human players. One recurring observation was that players occasionally chose to break partial sequences, even when alternative, unrelated cards were available for discard. We also saw a few moves where players discarded cards adjacent to the ones already picked by the next neighbor. This behavior is generally not expected and it was observed that in ~58% of such cases, the opponent picked the discarded card, indicating suboptimal decisions potentially driven by cognitive overload in later turns.

These examples highlight the dynamic and cognitively demanding nature of the game. We also observed instances where the model strongly preferred discarding duplicate cards, while players opted for alternative discards based on different inferred strategies. Overall, while the model aligns well with expert strategies in many scenarios, these deviations underscore the presence of nuanced, context-dependent decisions that merit deeper modeling and suggest promising directions for future refinement.

6.5 Game Semantics

6.5.1 Game Awareness: We examine the model's predictions in clear-cut scenarios to determine if it has learned typical game strategies. Table 8 shows these cases, the player action, and the model recommendation. It is evident how the model has learned basic game techniques such as avoiding breaking Pure Sequence (PS), avoiding discarding Joker, identifying and doing the valid declaration whenever possible etc., and its suggestions closely mimic those of highly skilled players.

Table 8. Model Predictions Reflecting Core Rummy Strategies. Cases where GNNetic replicates expert-level decisions demonstrating learned game awareness.

Case	Model Rec.
Breaks the Only Pure Sequence (PS)	0.02%
Discard Joker	0.10%
Breaks Impure Sequence when already has a PS	0.02%
Valid Declaration Possible	99.90%

6.5.2 Avoid Elimination: In Pool games, players play till they reach a particular score (61/101/201), depending upon the game type. If they reach or cross that score, they are eliminated. Towards the end of the game, when their score nears the elimination threshold, players prioritize survival over other factors. Analysis of HSP hands reveals that they identify and execute a discard that prevents elimination in 99% (Table 2) of cases wherever possible.

Table 9. Avoiding Elimination: A strategic discard choice where the player avoids crossing the 101-point threshold in a Pool-101 game.

Player Cards	(3♥, 4♥, 5♥, 6♥), (8♠, 9♠, 10♠, 11♠), (1♥, 1♣, 1♠), 5♦, 5♠, 2♦
Current Hand Score (14 cards)	12
Player Score	
Start of deal	92
If discarding 5♦/5♠	92+(12-5) = 99 (SAFE)
If discarding 2♦	92+(12-2) = 102 (ELIMINATE)

Table 9 shows a scenario in which a player in a pool-101 game begins a deal with 92 points. The player's best grouping of cards results in a hand of 12 points, bringing the total to 104 points. This is above the elimination threshold of 101 points, meaning the player would be eliminated from the game. Discarding the 2♦ would only reduce the player's score to 102, which is still above the cutoff. However, discarding either the 5♦ or the 5♠ would bring the player's score to 99, allowing them to avoid elimination, even though these cards form a partial meld. The model demonstrated its grasp of this strategy by recommending the same action, with these cards being the top predictions.

6.5.3 Player Last Deal: Players typically aim to maximize meld formation and minimize hand points. This strategy can change depending on the game type and how far along the game is. Players often discard high-value cards to lower their points as the game progresses. However, there are certain times when it is more beneficial to keep cards that could form melds in upcoming rounds rather than only optimizing for points. For example, in Table 10,

Table 10. Player Last Deal: Discard decisions in a high-risk final turn scenario, where any high-point discard leads to elimination.

Player Cards	9♥, 10♥, 3♦, 4♦, *A*♣, 2♣, 4♣, *JOKER*, 6♦, 6♥, 6♠, *J*♦, *Q*♦, 2♥
Current Hand Score (14 cards)	82 (max = 80)
Player Score	
Start of deal	96
If discarding J♦/Q♦	96+(82-10) = 168 (ELIMINATE)
If discarding 2♥	96+(82-2) = 176 (ELIMINATE)

a player began a deal with 96 points. Even with the best possible grouping of cards, the player still had 80 points in hand, resulting in 176 points – well over the 101-point limit for elimination. Instead of discarding high-value cards like J♦ or Q♦, the player chose to discard 2♥. This was because all the other cards were part of potential melds with other cards in hand. The model accurately predicted this move, demonstrating that prioritizing Pure Sequences can be more important than simply reducing points.

6.6 Game Outcome

To further evaluate the effectiveness of **GNNetic**, we analyzed how **playing skill**—as captured by discard behavior—relates to game outcomes. For this, we introduce a metric called **Discard Conformance**, which measures the percentage of a player's discard actions that matched the top-1 expert-recommended discard predicted by GNNetic. For evaluating game outcomes, we used Win Rate, defined for Pool games as the ratio of games won to games played. For each player, we calculated their overall discard conformance on all historical games played by them, along with the number of games they won. For each discard conformance bucket, we calculated the average win rate across all the players falling in that conformance range. This indicates the distribution of players across different discard conformance levels. We can see that the player's win rate increases with an increase in Discard Conformance. A similar trend is observed in reverse for the End Score (average number of points at the end of the deal), where improvement in Discard Conformance leads to improvement in the End Score (Fig. 9 shows for Pool-6P games).

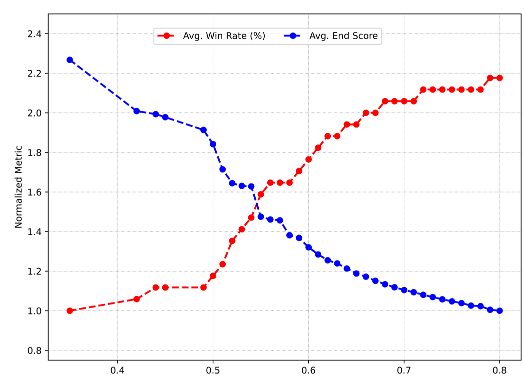

Fig. 9. Discard Conformance vs. Win Rate and End Score (Pool-101, 6P). Higher conformance aligns with better outcomes—higher win rates and lower end scores. This supports the HCI goal of using feedback and cognitive scaffolds to enhance decision-making in complex games.

6.7 Evaluation Across Rummy Variants (Points and Deals)

Table 11. Accuracy Comparison of CardNet and GNNetic on different Rummy Variants

Accuracy	Points			Deals		
	CardNet (%)	GNNetic (%)	Gain (%)	CardNet (%)	GNNetic (%)	Gain (%)
Top-1	66.58	67.15	**0.85**	58.33	64.44	**10.47**
Top-2	85.96	86.40	**0.51**	78.78	83.25	**5.67**
Top-3	93.72	94.23	**0.54**	88.37	91.28	**3.29**
Threshold	81.25	82.88	**2.01**	72.23	81.72	**13.14**

To evaluate GNNetic's generalizability, we extended our experiments to other popular Rummy variants, namely *Points* Rummy and *Deals* Rummy. For these variants, the hand-related features remained identical to those listed in Table 1, while the game-context features were adapted to capture variant-specific dynamics. For example, in Deal Rummy, which runs for a fixed number of deals, we introduced features such as *deal_num* (current deal number) and *user_rank* (the player's standing at the start of the deal). For Points Rummy, which is structurally simpler, we used only the relevant card, hand, and game-level attributes mentioned in Table 1. Any features specific to Pool-varaint (like *Can be eliminated*, *Current Game Score Ratio*, *Game Variant* etc.) were discarded.

For training, the graph architecture and the methodology was kept the same as outlined in Section 5.2 (Fig. 3) and 5.4 respectively. Accuracy was then measured for both the models - CardNet and GNNetic. GNNetic demonstrated varying degrees of improvement across these variants. The percentage gain over CardNet [18] was relatively modest in Points Rummy, primarily because this format is simpler and outcomes are driven predominantly by the cards in play, with minimal influence from broader game context. In contrast, the gain in Deals Rummy was substantially higher. Player discard behavior in Deals is strongly influenced by contextual factors such as ranking, deal number, and the points accumulated before the start of each deal. By effectively leveraging these variant-specific contextual features, GNNetic achieved a significant performance boost, highlighting its ability to capture deeper game dynamics for improved prediction accuracy. Consistent with results on the Pool variant, GNNetic outperformed CNN-based models, delivering superior accuracy while maintaining a significantly smaller model size, underscoring both its efficiency and adaptability across Rummy formats. (Table 11).

Importantly, this cross-variant robustness ensures that the insights provided by GNNetic remain consistent and interpretable, allowing features like Companion Mode and Skill Arena to deliver reliable and variant-aware interface and player insights respectively for different variants of Card Games.

7 Player-Centered Explainability in GNNetic

While earlier sections discussed and demonstrated GNNetic's predictive accuracy in deeper details and diverse angles, this alone is insufficient for our broader goal: helping players improve and informing design interventions. In skill-based games like Rummy, explainability is essential too—not just for trust in predictions, but to generate feedback that is actionable, intuitive, and meaningful to both players and designers.

Generic post-hoc GNN explanation methods such as GNNExplainer [45] and GraphLIME [27] reveal node- or edge-level contributions, but they offer abstract insights better suited to model developers than end-users. GNNetic instead employs a player-centered explainability approach, surfacing rationales through domain-relevant constructs like meld formation, joker use, hand minimization, and valid declarations.

Given a hand and declared joker, GNNetic's explainability module outputs the best possible grouping, along with the meld types (e.g., pure/impure sequences, sets, hand points, valid declaration) using an adapted implementation of the Best Grouping Formation algorithm described in [15, 16]. On combining this output with the predictions from GNNetic, it allows the system to identify recurring suboptimal patterns—such as players discarding jokers that could have completed melds or struggling to form valid declarations despite viable groupings or making invalid declarations due to confusion in grouping logic. These behaviors are then examined in conjunction with game context (e.g., hand layout, card positioning, joker visibility, prior actions) to infer their likely source.

- If such errors occur frequently across a broad player base, especially around the same UI components (e.g., unclear joker indicators, lack of visual meld grouping cues), they are indicative of **UI shortcomings**. For instance, if players repeatedly discard jokers placed at the edge of the hand view or fail to detect meld opportunities despite correct logic, the design may not be drawing sufficient attention to critical elements.

Fig. 10. With auto-sorting, cards are automatically grouped by their suit type and rank on the first turn of the player, making it easier for players to quickly understand and evaluate the dealt hand.

- Conversely, if these patterns are concentrated among newer or lower-performing players and diminish with experience, they are more likely due to **skill gaps**. In such cases, the same interface is available to all, but only some users make consistent mistakes—pointing toward the need for scaffolded learning (e.g., better tutorials, post-game tips).

By combining GNNetic's best grouping output with player decision trails and interface telemetry, the system can help isolate whether the breakdown lies in perception (UI issue), comprehension (skill gap), or strategic execution (lack of experience). By translating model outputs into visual, intuitive, and semantically aligned explanations, GNNetic supports both metacognitive reflection and interface refinement, aligning with HCI goals of reducing cognitive load and improving strategic decision-making. These explainability outputs directly inform the design of two key intervention modes—**Companion Mode** and the proposed **Skill Arena**—which we describe in the following sections.

8 Companion Mode: Leveraging Insights for UX Enhancement

To extend the utility of the discard prediction model beyond mere prediction, we leveraged insights generated by GNNetic to proactively identify patterns of player error and inform design improvements grounded in Human-Computer Interaction (HCI) principles. By systematically comparing GNNetic's model-recommended optimal discards with actual player choices, we surfaced behavioral divergences. For instance, our analysis of scenarios leading to invalid declarations revealed player struggles in building correct melds while examinations of joker discards highlighted oversights despite players often exhibiting good judgment in discarding other cards. This analysis confirmed that many of these identified mistakes did not stem from randomness or a general skill gap, but rather from systematic decision biases and potential shortcomings in the interface's ability to support informed play. From an HCI perspective, these findings highlighted specific opportunities to scaffold player decisions through more intuitive, informative, and context-sensitive UI design.

Based on these insights, we designed and implemented a set of UX interventions aimed at gently guiding players toward more strategic discard choices in practice games. These included:

- Automatic Sorting: Cards are now automatically sorted upon game start (grouped by suit and sorted by rank), helping players save time and focus on strategy (Fig. 10).

Fig. 11. Different Melds (like Sequence, Set, Pure Sequence, Invalid), Joker tag being marked along with Meld Score cues in Companion Mode, thereby reducing Cognition load from the user.

- Meld Marking and Scoring: Visual highlighting of player-formed melds, along with their associated scores, was introduced to reduce the cognitive burden of hand evaluation (Fig. 11).
- Joker Card Marking: Addressing observed instances where players accidentally discarded Jokers due to oversight rather than strategy, the designated Joker card is now distinctly marked with a "J" tag ($K\spadesuit$ in Fig. 11). This enhances its visibility and prevents erroneous discards.

These interventions reflect broader learning design principles. By embedding task-relevant cues directly within the gameplay interface in practice games, Companion Mode provides **situated scaffolding** [10] - a form of support that helps players perform complex tasks by reducing extraneous decision load in context. In doing so, the interface also adheres to the principles of **ecological interface design** [42], promoting usability by aligning visual affordances with core cognitive demands of the task.

8.1 Impact Evaluation

Following these changes, we measured the impact on player behaviour. To ensure fairness and consistency, all the design changes were rolled out to all the players at once, thus making A/B experimentation impossible, and we relied on pre-post-comparative analysis.

In our experiment, we compared player behaviour for 1 Month before (hereby referred to as pre-period) and after launching the changes (hereby referred to as post-period). Both groups had 3-4 million player games as part of the study.

As shown in Table 12, after the introduction of Companion Mode in practice games, the Joker discard percentage reduced significantly. We further evaluated the discard of Ace cards when the printed joker was the declared wild joker (Special Case: Ace becomes wild card joker as per Rummy rules). We observed a substantial improvement in Ace joker discard behavior among players using Companion Mode, with a reduction of **48%**. For a fair comparison, we also measured Ace joker discard in non-practice games (where Companion Mode wasn't offered) during the same period. We didn't see any relatively substantial reduction (4%) in Ace joker discard for such cases. This could either be attributed to randomness or players now knowing better what the implicit joker card is when the declared joker is a printed joker.

Table 12. Impact of Companion Mode on player behavior - pre and post-deployment comparison in practice games. Post-deployment analysis shows significant drops in joker discards (31%) and invalid declarations (27%), highlighting the efficacy of targeted UI changes informed by GNNetic insights.

Player Action	**Game-Type**	**Pre-Period**	**Post-Period**	**Reduction**
Joker Discard *(as % of total moves)*	with Companion Mode	0.420%	0.320%	31.25%
	without Companion Mode	0.149%	0.144%	3.36%
Joker Discard (when printed Joker is joker) *(as % of total moves)*	with Companion Mode	0.960%	0.647%	48%
	without Companion Mode	0.246%	0.237%	4%
Invalid Declaration *(as % of total games)*	with Companion Mode	0.47%	0.37%	27.03%
	without Companion Mode	0.51%	0.50%	2.00%

Beyond specific card discards, these design interventions also aimed to improve declaration accuracy. Our analysis showed a measurable reduction in invalid declarations, with an improvement of **27%**. This was driven by the meld marking and meld scoring. This further highlights how improved visual affordances directly mitigate errors related to complex rule application and cognitive load during critical gameplay moments. For other non-practice games, this number again showed a non-substantial reduction (2%) in invalid declarations.

These findings demonstrate how model-informed UI design can improve decision-making by reducing cognitive load and surfacing critical cues in practice games. While Companion Mode does not adapt in real time, its design was shaped through data-driven iteration, showing how predictive models like GNNetic can inform effective interface scaffolding for supporting strategic play in complex multiplayer games.

8.2 Summative User Study: Companion Mode Feature Impact and Design Implications

To further understand the impact and utility of our newly rolled out *Companion Mode* features on player experience and cognitive load, we conducted a summative user study. This aimed to evaluate if and how these features helped players, through non-leading questions that encouraged genuine feedback.

Methodology: We recruited players from three distinct cohorts for structured interviews, collecting a total of 83 complete responses:

- Cohort 1 (n=31): Newer players who just played 3-5 games.
- Cohort 2 (n=25): Players who played 10-15 games and demonstrated high skill scores.
- Cohort 3 (n=27): Recent highly active players (games played > 20).

Interviews focused on perceived differences in game table design, comparing the pre-launch (legacy) gametable with the post-launch gametable featuring Companion Mode, and explored the utility of observed assist features. Here are some of the questions:

- Did you notice anything different on the game table after the recent update?
- Can you describe your experience playing a game with the updated table design?
- Were there any specific aspects of the game table that helped you with your decisions?

Key Findings and Design Implications: A significant majority across all cohorts (71-76%) reported noticing differences on the game table compared to our pre-launch design, affirming the demand for specific assist features. The most frequently cited and highly valued features were:

- **Auto Sort (25%-30% across cohorts):** Players consistently highlighted this feature's ability to save time by automatically arranging cards, allowing them to focus on game strategy and aiding decisions on their first turn (e.g., whether to drop a deal).
- **Joker Marking (16%-21% across cohorts):** This feature was valued for saving time in quickly identifying the Joker card, facilitating its use in meld formation, and crucially, preventing accidental discards of the Joker.
- **Melds Score / Total Points Display (16%-25% across cohorts):** Players utilized this for risk assessment to decide whether to continue a deal and to maintain a continuous track of their points across multiple games.
- **Sequence/Set Indicators (11% in Cohort 2):** Providing visual confirmation of valid sequences and sets directly helped players avoid incorrect declarations and quickly verify their meld formations.

These findings directly informed our iterative UI refinements, highlighting critical areas where visual affordances and automated assists could significantly reduce players' cognitive load and enhance usability. The consistently high utility reported for features preventing common errors (e.g., accidental Joker discard, invalid declarations) and supporting complex hand evaluation underscores the opportunity for design to directly support player experience and accelerate skill acquisition in Online Rummy.

9 Skill Arena: Post-game Insights

Beyond informing interface design, GNNetic's predictive capabilities are extended to support personalized skill development through a post-game feedback feature called Skill Arena, focused on discard decisions. This builds on our earlier work—SPADENet [21, 28] - a model trained on expert player behavior to determine whether a player should do *First Drop*, a critical early-game decision to minimize losses from weak hands.

To operationalize this, we launched a feature that surfaces Drop or Play recommendations on the post-game scorecard, enabling players toward more strategic decisions. Each insight includes three core components: the player's actual action, the expert-modeled recommendation, and the expert action outcome with a concise explanation grounded in strategic reasoning (e.g., hand strength too low, no viable melds, points saved). These recommendations are designed to be brief, visually accessible, and context-aware, aligning with players' limited post-game attention and supporting reflection-in-action (Fig. 12).

To enhance accessibility and inclusivity, insights were localized into multiple Indian languages (e.g., Hindi, Tamil, Kannada), enabling players to consume content in their preferred language. For sustained engagement, we introduced a Game History section (Fig. 13), where players can revisit past feedback, not just for First Drop, but also for declaration accuracy and win-rate trends across Rummy formats. To reinforce continuous learning, we introduced a Drop Score—a dynamic, gamified metric that tracks alignment with expert-modeled drop decisions. This supports self-regulated

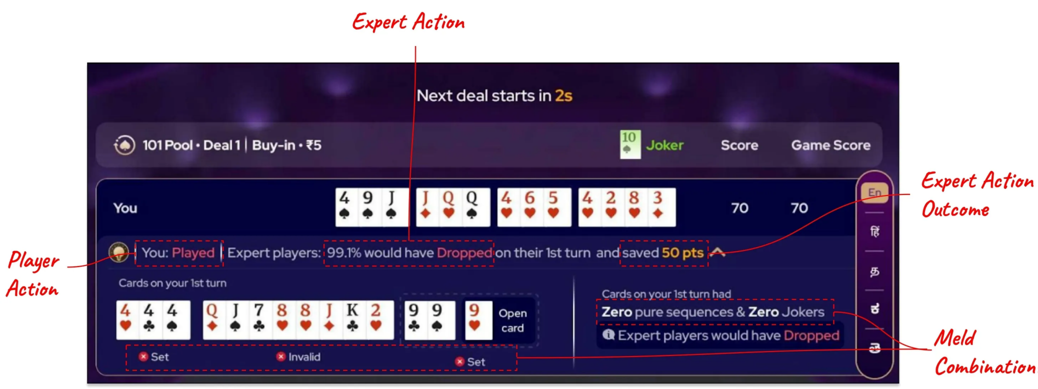

Fig. 12. Drop or Play Insights on the game scorecard after the end of each game based on expert player action. Players can customize the language based on their preference.

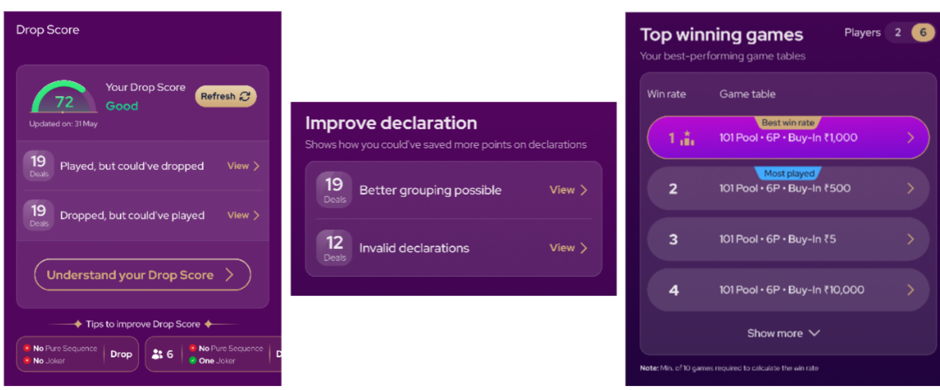

Fig. 13. Insights Section where Players can review their past games and drop score. The score is dynamic and players can monitor how it changes with each game they play. Along with that, players can also get other insights related to declaration and their historical win-rates in different formats.

learning, encourages mastery through goal-setting and progress tracking, and creates a sense of achievement through immediate, interpretable feedback. From an HCI standpoint, Skill Arena exemplifies scaffolded feedback design, providing learners with reflective, actionable insights that are grounded in gameplay context and presented in a cognitively manageable format.

Impact of Drop Insights on Player learning: To evaluate the effectiveness of post-game Drop or Play insights, we conducted an A/B experiment in which 50% of players were exposed to these insights via the scorecard and Game History, while the remaining players saw no changes. The results showed a significant improvement, particularly among error-prone players in the *Test Group*. Those who previously struggled with First Drop decisions improved their Drop Conformance by up to **4x** after engaging with the feedback. Players in the test-group who reviewed past mistakes more frequently (10+ times) showed a **2–2.5x** improvement compared to their control-group peers.

Fig. 14. Post-game Scorecard with summary of discard deviations–prompting reflection by quantifying key mistakes. A deep-link to detailed Skill Arena insights supports self-regulated learning without disrupting the game's flow.

From a Human–Computer Interaction (HCI) perspective, the feature addresses key principles of interpretability and user-centered design in skill-based gaming. By translating complex model outputs into clear, contextual, and language-accessible feedback, the system reduces cognitive load and supports reflection—bridging the gap between novice and expert decision-making.

These findings demonstrate that players benefit significantly from structured, post-game insights, especially in high-load decision environments like Rummy. The marked improvement in Drop Conformance highlights a strong appetite for actionable, strategic cues. Building on this success, we now extend these insights to discard decisions through the proposed Skill Arena—a post-game interface that offers personalized, discard-level insights based on GNNetic's expert predictions. While evaluation of this new feature is ongoing, its design is grounded in proven mechanisms that enable metacognitive reflection and strategic learning.

The following subsections detail how discard-level insights are surfaced through Skill Arena, with a focus on interface design and delivery strategies that aim to maximize clarity, engagement, and learning efficacy.

9.1 Discard Insight Summary on Scorecard

At the end of every game, players see a summary of discard tips on the scorecard screen—highlighting the total number of moves where they deviated from GNNetic's expert recommendation. Next to this summary, players are given the option to send themselves a deep link to review the insights later. This is designed to support delayed reflection, a key principle in self-regulated learning, by allowing players to revisit key moments without being overwhelmed immediately after the game. Fig. 14 shows this scorecard interface, and Fig. 15 illustrates how players access detailed tips via the Skill Arena or the deep link.

Once in the *Skill Arena* post their gameplay, players can systematically review their deviations, based on mistakes identified by GNNetic's expert recommendations. These insights will present the best possible action and context, accompanied by detailed explanations that incorporate all game table elements players consider (e.g., meld marking, joker identification, current hand points, etc.). Players will see their 14-card hand and respective hand points value, along with a visual

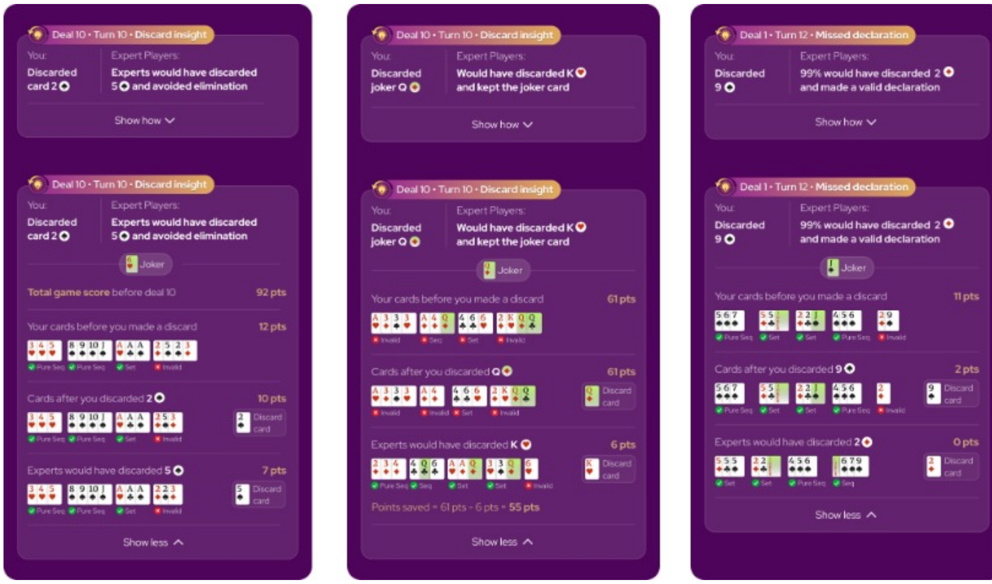

Fig. 15. Skill Arena interface displaying post-game discard insights–empowering players to retrospectively examine key discard mistakes by comparing their chosen action with expert-model recommendations. Visual overlays explain the impact of each decision (e.g., point change, meld structure), fostering reflection and strategic improvement without influencing live gameplay.

representation of how their hand was impacted by the card they discarded. For direct comparison, they'll also see the state of their hand had they followed the expert model's recommendation. This comparative visualization will enable players to clearly discern the superior expert choice and internalize the reasoning. For example, in a Pool-101 game depicted in Fig. 15, a player at 92 points discarded a 2♠, reducing their hand from 12 to 10 points. If an opponent declared, this player would be eliminated, exceeding the 101-point threshold (92+10 = 102). The "Experts' Action" visual part clearly highlights how an expert discard (5♠) would have kept the player's total points below the 101 threshold, thus remaining in the game. This direct visual comparison helps players quickly grasp the impact of their decision and assimilate the learning.

We plan to show such tips for critical error types, including discarded jokers, invalid declarations, missed opportunities to prevent elimination, discards that break melds (leading to increase in hand points), or discarding a card adjacent to one picked by a neighbor etc. This reflective analytics interface is designed to foster metacognitive awareness of their decision-making processes, providing concrete examples of suboptimal plays. This approach reflects principles of deliberate practice [17], where learners improve by reviewing targeted feedback in areas just beyond their current competence. The structured nature of these post-game reviews also aligns with scaffolded learning [20], supporting improvement within the player's Zone of Proximal Development (ZPD). Over time, repeated exposure to such reflective analysis may aid in the development of domain-specific mental models and pattern recognition abilities, which are hallmarks of expertise development in complex environments [9].

By grounding feedback in the player's actual gameplay decisions and framing it within intuitive visual comparisons, the Skill Arena promotes metacognitive awareness—helping players better monitor, evaluate, and refine their strategies. In this way, GNNetic is not only a predictive engine, but also a cognitive support tool that fosters skill growth through structured reflection and informed self-assessment.

10 Conclusion

This work introduces GNNetic, a novel Relational Graph Convolutional Network that bridges advanced AI analysis with crucial HCI insights in complex multiplayer card games. By comprehensively modeling player decision-making through a novel graph-based representation of card interactions and game-level context, GNNetic serves as a powerful framework for understanding and enhancing player interaction. With Online Rummy as a Case Study, our findings demonstrate GNNetic's technical strengths: it achieves better accuracy while being 15 times smaller than existing models. Crucially, Label Smoothing ensures GNNetic accurately identifies all valid discard options, making it well-suited for diagnostic and educational feedback. GNNetic's primary contribution lies in its application to cognitively-informed game design and skill development. Our summative study leveraged GNNetic's predictions to identify common decision bottlenecks, informing interface enhancements such as joker highlighting, meld marking, and auto-sorting. These changes led to measurable improvements in practice-game behavior (~31% fewer joker discards; ~27% fewer invalid declarations), demonstrating how UI refinements guided by gameplay analytics can reduce extraneous cognitive load. Moreover, GNNetic's core representation generalizes effectively across popular Rummy variants, such as Pool, Deal, and Points Rummy, by incorporating variant-specific features, further validating its adaptability and broader applicability.

For individual learning, GNNetic powers the proposed Skill Arena - a post-game feedback interface surfacing discard-level deviations. This design builds on our prior work with First Drop insights, where post-game feedback improved Drop Conformance by **2–2.5x**. By enabling players to reflect on missed cues and better understand expert reasoning post-game, Skill Arena fosters structured reflection and metacognitive support—key pillars of expertise development. Together, these tools lay the groundwork for a scalable, data-informed tutoring framework in multi-player gaming environments, advancing both usability and fair skill enhancement.

11 Future Work

GNNetic opens promising directions for extending both its analytical capabilities and its impact on human–computer interaction. While the current model demonstrates strong performance in discard prediction, card picking remains another distinct and critical decision point in Rummy. Just like discards, this action can be modeled using GNNetic's relational architecture—predicting whether a player should draw from the discard pile or the closed deck based on hand composition, visible discards, and other contextual cues. Expanding GNNetic to jointly model both discard and pick decisions would enable a more holistic representation of player strategy, strengthening its value for both real-time game analytics and post-game metacognitive feedback.

Beyond Rummy, the architecture lends itself to other imperfect-information multiplayer card games, such as Poker. Here, individual cards (hole and community) can be encoded as nodes with edge types like SAME-SUIT, SAME-RANK, and SEQUENTIAL-RANK to model key hand structures. Game progression across betting rounds (pre-flop, flop, turn, river) can be represented via evolving graph structures or temporal graph layers. Player-specific features like position, stack size, and betting history can enrich the node features, enabling GNNetic to learn patterns related to bluffing, folding, or hand strength estimation. To explicitly address strategic deception, historical behavioral data (e.g., frequency of raises, folds, or bluffs) can be sourced from HUD statistics or longitudinal player behavior logs and incorporated into the graph as opponent modeling features, allowing the model to reason about non-rational and adversarial tendencies in gameplay. Similarly, in MTG-like games, where cards have diverse modalities (creatures, spells, effects), nodes can represent cards with attributes encoded as features, and edges can encode interactions (e.g., TARGETS, BUFFS, SYNERGIZES-WITH). These parallels suggest GNNetic's design is extensible to broader domains with relational complexity and partial observability.

Beyond predictive accuracy and gaming applications, this work emphasizes interpretability from HCI perspective, surfacing player-centric discard tips and associated reasoning for immediate usability in skill development and interface enhancements. While from a technical aspect, edge ablation study has been done to understand the impact of relationship over overall discard decisions, further study can be done to understand per-feature importance for individual predictions. Methods like Integrated Gradients [38] and GraphLIME [27] can be explored.

12 Broader Impact

From an HCI and Learning Evolution perspective, future work will focus on advancing the "Skill Arena" into a fully adaptive post-game learning system. This system would dynamically tailor recommendations and learning pathways based not only on a player's historical decision trails but also on latent behavior embeddings learned by GNNetic. These embeddings can serve as a basis for player segmentation from their playing behavior perspective via unsupervised methods (e.g., clustering), revealing groups with similar skill levels, learning patterns, or strategic preferences. Instructional content could then be personalized by both learning style (e.g., visual learners receiving animated replays; active learners engaging in what-if simulations) and inferred mastery (e.g., basic meld guidance for beginners, discard optimization for advanced users). Additionally, the system could scaffold advanced strategic concepts such as risk management, information exploitation, opponent modeling, and resource allocation, contextualized to a player's decision-making patterns. By combining data-driven player modeling with metacognitive support, Skill Arena can evolve into a scalable, personalized tutoring system for strategic upskilling in card games.

Furthermore, we aim to explore explicit links between transferable cognitive skills fostered in games and their applicability to real-life domains. For instance, assessing incomplete information in Rummy (discerning opponent's hands from discards) directly translates to evaluating market trends with limited data; managing probabilistic outcomes (weighing odds of drawing a card) applies to strategic financial decisions or evaluating project success likelihood; and rapid pattern recognition (identifying melds or opponent's bluffing patterns) extends to quickly diagnosing equipment malfunctions or understanding social dynamics. By explicitly linking these in-game cognitive processes to analogous real-life challenges, we aim to demonstrate how engaging with strategically rich games can cultivate broadly applicable decision-making abilities.

13 Responsible Deployment and Ethical Reflection

While GNNetic's insights have clear pedagogical potential, we recognize the unique ethical sensitivities involved in sharing post-game insights in any multi-player gaming environments. First, to preserve fairness and game integrity, all discard insights and feedback are surfaced only after the game concludes, ensuring they do not influence live outcomes or decision-making. This separation reinforces the tool's usage as a learning aid rather than a real-time performance booster.

Second, to preserve player autonomy and consent, feedback interfaces like Skill Arena shall be designed to be accessed through opt-in, participatory flows. Players choose whether to review their own gameplay, and can revisit insights at their convenience - encouraging reflective, self-paced learning rather than imposed evaluation.

Third, confidentiality is strictly upheld. All feedback is shown only to the player who made the decisions. No individual playing patterns, mistakes, or strategic preferences are ever exposed to opponents or made public, preventing any competitive disadvantage or misuse of insights as a surveillance tool.

Finally, beyond serving as a source of entertainment, learning, or engagement—by adding an additional layer of challenge - such performance reflection tools also enable players to recognize common error patterns and refine their strategic thinking over time. By supporting more informed choices, these tools can enhance situational awareness and mitigate the effects of cognitive overload.

Glossary

Card A playing piece defined by a Rank (1 to 13) and a Suit (♦, ♠, ♣, ♥). 3

Closed Deck A shuffled pile of face-down cards left after dealing. Players draw blindly from this stack. 3

Deal A single round of play where each player is dealt 13 cards, and a wild joker is selected. Players take turns drawing and discarding cards until someone declares or the round ends otherwise. 3

Deals A multi-round version of Rummy where each player plays a fixed number of deals (e.g., 2 or 3). Players aim to minimize their total penalty points across all rounds. After the final deal, the player with the lowest cumulative score is declared the winner. 3, 9, 21

Deck A standard playing card collection. Most rummy variants are played with 2 decks, each having 53 cards (For each suit: 13 cards — 2-10, Ace, Jack, Queen, King; 4 suits — clubs, diamonds, hearts, spades; plus 1 Printed Joker). 3

Drop A move where a player voluntarily exits a deal to avoid losing more points. A *First Drop* occurs before the first move (lower penalty), while a *Middle Drop* happens after at least one move (higher penalty). 3, 9

First Drop First Drop refers to a player choosing to exit the game before making their first move, i.e., before drawing any card from the open or closed deck. 6, 11

Impure Sequence / Sequence A straight of 3 to 5 cards from the same suit where one or more cards are replaced by jokers. Example: (7♠, 5♦, 9♠, 10♠), where 5♦ acts as a joker. 3

Invalid Declaration When a player declares their hand but fails to meet the structural requirements of a Valid Declaration. 3

Joker Card A wildcard that players can use as a substitute for any card in their hand. It can be a printed joker card or a wild card joker. 2, 3

Meld A general term for any of the following combinations: Pure Sequence, Impure Sequence, Set, or Invalid. 2, 3

Open Deck Also called the Discard Pile; a stack of face-up cards from which players can pick during their turn. Visible to all players and provides strategic information. 3

Partial Meld An incomplete group of cards with the potential to form a valid Pure Sequence, Impure Sequence, or Set, but not yet meeting the criteria.
Examples:
(5♠, 6♠): can become a Pure/Impure Sequence with 4♠, 7♠ or a joker.
(K♦, K♣): needs another King of a different suit or joker to complete a Set.
(7♦, 9♦): adding 8♦ or a joker completes a Sequence. 13

Pick An action where a player draws one card at the start of their turn—either from the Closed Deck (face-down, unknown card) or the Open Discard Pile (face-up, previously discarded card). This decision is strategic, often influenced by the player's current hand and inferred intentions of opponents. 9

Points A quick, single-deal version of Rummy where players aim to make a valid declaration as early as possible. Each card has a predefined point value, and the winner scores zero while others accumulate penalty points based on unmatched cards in hand. The winner's score is calculated based on the total points of the opponents. 3, 9, 21

Printed Joker Card A printed joker is a card with a unique "joker" design and can substitute for any other card when forming melds. There is only one such card in each deck. 3

Pure Sequence A straight of 3 to 5 cards in sequential order, all sharing the same suit. Example: (3♠, 4♠, 5♠). 2, 3

Set A group of 3 or 4 cards that share the same rank but come from different suits. Example: (7♦, 7♠, 7♣, 7♥). 2, 3

Valid Declaration A hand that includes at least two Sequences, one of which must be a Pure Sequence, and contains no Invalid melds. Remaining combinations can be any valid meld. 3

Wild Card Joker A randomly selected card (e.g., 7♦ at the start of each deal, which acts as a joker in that particular game. All cards of the same rank but any suit (e.g., 7♦, 7♠, 7♣, 7♥) function as jokers for that round. 3

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